

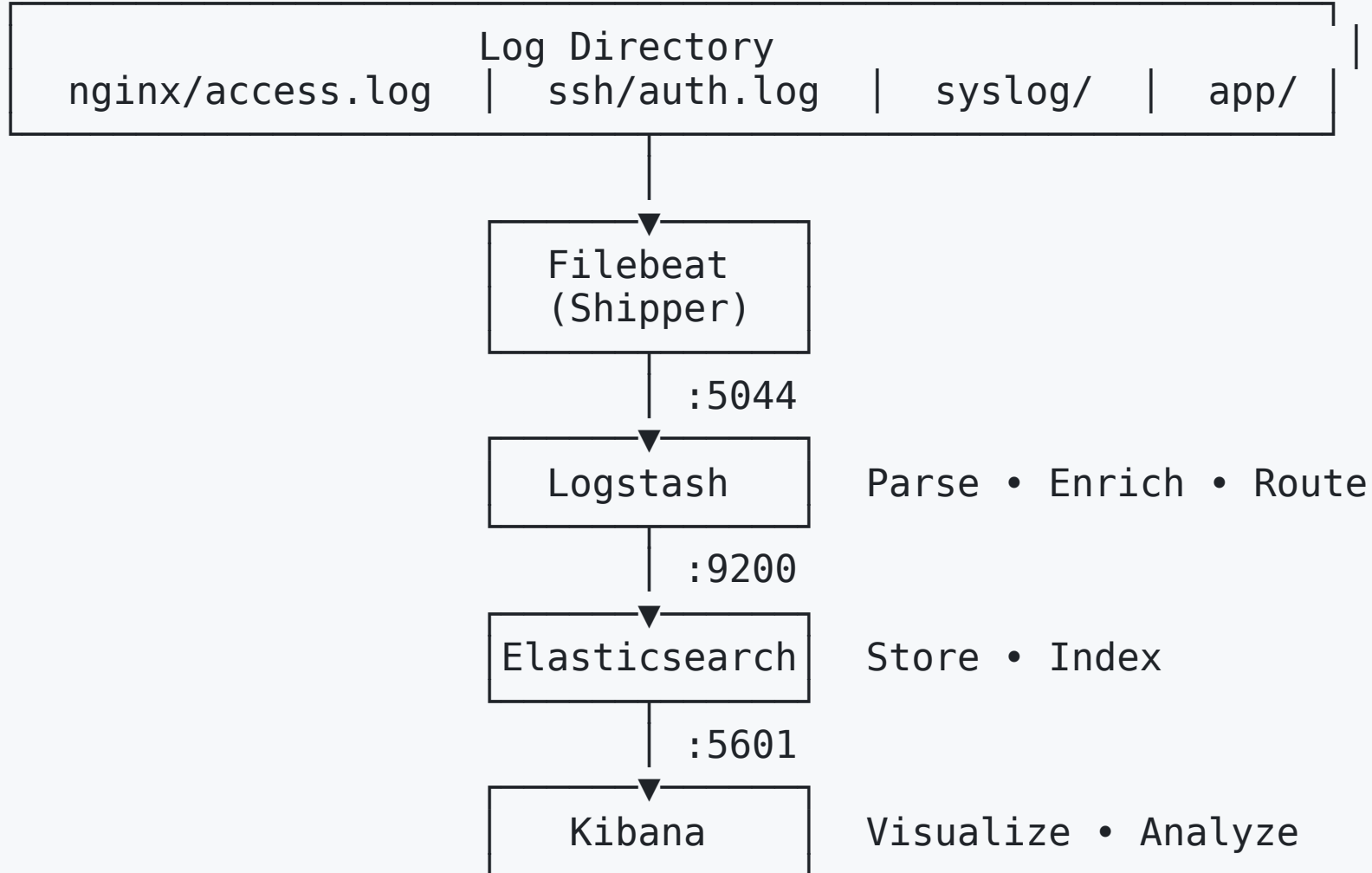
ELK Lab: Observability & Centralized Logging

Systems Reliability Engineering (SRE)

Agenda

- Architecture Overview
- The 5 Log Sources
- Setting Up the Stack
- Configuring Kibana
- Generating Test Data
- Learning Outcomes

Architecture Overview



The 5 Log Sources

Source	Format	Key Fields	Use Case
Nginx	Combined Apache Log	clientip, status, request	Web traffic analysis
SSH	JSON	user, result, source_ip, geoip	Security monitoring
Syslog (Linux)	RFC5424	hostname, process, message	System events
Syslog (Cisco)	CEF	src_ip, dst_ip, severity	Network device logs
App Service	JSON	method, endpoint, status_code, latency	Application health

Log Source: Nginx

Format: Combined Apache Log

```
192.168.1.50 - - [05/May/2024:14:30:00 +0000] "GET /checkout HTTP/1.1" 200 1234 "-" "Mozilla/5.0"
```

Enriched Fields:

- `clientip` → GeolP lookup (country, location)
- `timestamp` → Parsed via date filter
- `status` → HTTP status code
- `request` → URL path and method

Log Source: SSH

Format: JSON (structured)

```
{  
  "timestamp": "2024-05-05T14:30:00Z",  
  "event": "ssh_auth",  
  "user": "admin",  
  "source_ip": "185.234.72.45",  
  "result": "failure"  
}
```

Key Insight: 80% successful logins (internal IPs) vs 20% failures (external attack IPs)

Log Source: Syslog (Linux)

Format: RFC5424

```
<34>1 2024-05-05T14:30:00Z webserver ssh 1234 - - Failed password for invalid user admin from 185.234.72.45 port 54321 ssh2
```

Parsed Fields:

- `priority` → severity + facility
- `hostname` → origin system
- `process` → service name
- `message` → the actual event

Log Source: Syslog (Cisco)

Format: CEF (Common Event Format)

```
CEF:0|Cisco|IOS|12.4|5|SSH login attempt|3|src=192.168.1.100 dst=10.0.0.5 spt=54321 dpt=22
```

Fields: Device type, severity level, source/destination IPs, ports

Log Source: App Service

Format: JSON (structured)

```
{  
  "timestamp": "2024-05-05T14:30:00Z",  
  "method": "POST",  
  "endpoint": "/api/checkout",  
  "status_code": 200,  
  "latency_ms": 145  
}
```

Metrics Tracked: Response times, error rates, endpoint popularity

Setting Up: Start the Stack

Step 1: Launch all services

```
cd /home/morta/workspace/elk-lab/docker  
docker compose up -d
```

Step 2: Wait for initialization (30-60 seconds)

```
docker compose ps
```

Step 3: Verify services are running

```
curl http://localhost:9200          # Elasticsearch  
curl http://localhost:5601/api/status # Kibana
```

Accessing Kibana

Open in browser:

```
http://localhost:5601
```

First Time Setup:

1. Click "Explore on my own" (skip tutorial)
2. Stack Management → Index Patterns
3. Create patterns for each log source

Configuring Kibana: Index Patterns

Create these patterns in order:

Pattern	Time Field
logs-nginx-*	@timestamp
logs-ssh-*	@timestamp
logs-syslog-*	@timestamp
logs-cisco-*	@timestamp
logs-app-*	@timestamp

Note: You can also use `logs-*` to query all sources at once

Configuring Kibana: Create Visualizations

Recommended Visualizations:

1. **SSH Attack Map** (Maps)

- Layer: Terms on `geoip.location`
- Source: `logs-ssh-*`

2. **SSH Auth Results** (Pie Chart)

- Slice by: Terms → `ssh_result`

3. **Nginx Traffic Over Time** (Line Chart)

- X-axis: Date histogram → `@timestamp`

4. **HTTP Status Codes** (Bar Chart)

- X-axis: Terms → `status`

Build a Dashboard

Steps:

1. Click **Dashboard** → **Create dashboard**
2. Add all saved visualizations
3. Arrange and resize panels
4. Save as "ELK Lab Overview"

Pro Tip: Use `logs-*` pattern to see all data, or filter by specific source using the `service` field

Generating Test Data

Nginx Logs (web traffic)

```
curl http://localhost:80  
curl http://localhost:80/products/123  
curl http://localhost:80/checkout
```

SSH Logs (security events)

```
docker exec ssh-simulator /generate_ssh_logs.sh 50  
# Generates 80% success (internal) + 20% failure (external)
```

Generating Test Data (continued)

Syslog Events

```
docker exec syslog-simulator /generate_syslog.sh 100  
# 50 Linux RFC5424 + 50 Cisco CEF events
```

App Logs (API requests)

```
curl http://localhost:5000/api/health  
curl -X POST http://localhost:5000/api/checkout \  
  -H "Content-Type: application/json" \  
  -d '{"item":"test","user_id":"user-123}"'
```

Verifying Data Arrived

In Kibana Discover:

1. Click **Discover** (left sidebar)
2. Select `logs-*` index pattern
3. Set time range to "Last 15 minutes"
4. You should see documents appearing

Via Command Line:

```
curl localhost:9200/_cat/indices?v
```

Look for indices like: `logs-nginx-2024.05.05`

Logstash Pipeline: How It Works

```
input { beats { port => 5044 } }

filter {
  # Route by service type
  if [service] == "nginx" { grok { ... } geoip { ... } }
  if [service] == "ssh"   { json { ... } geoip { ... } }
  if [service] == "syslog" { grok { ... } }
  if [service] == "cisco"  { grok { ... } }
  if [service] == "app"    { json { ... } }
}

output {
  elasticsearch { index => "logs-%{[service]}-%{+YYYY.MM.dd}" }
}
```

Key: Dynamic index naming by service + date

Learning Outcomes

By completing this lab, you will be able to:

1. **Configure log shippers** - Set up Filebeat to read multiple sources
2. **Parse unstructured logs** - Use Grok filters for field extraction
3. **Enrich telemetry** - Add GeoIP for geographic context
4. **Manage indices** - Create daily indices per log source
5. **Build dashboards** - Visualize correlated telemetry
6. **Analyze security events** - Identify brute-force attacks via SSH + GeoIP
7. **Monitor app health** - Track latency and error rates

Key Commands Reference

Start/stop

```
docker compose up -d
```

```
docker compose down
```

View logs

```
docker compose logs -f filebeat
```

```
docker compose logs -f logstash
```

Test Elasticsearch

```
curl localhost:9200/_cat/indices?v
```

Generate data

```
docker exec ssh-simulator /generate_ssh_logs.sh 50
```

```
docker exec syslog-simulator /generate_syslog.sh 100
```

Restart a service

```
docker compose restart filebeat
```

Troubleshooting

No data in Kibana?

```
# Check Filebeat
docker exec filebeat filebeat test config
docker logs filebeat

# Check Logstash
docker compose logs logstash | tail -50

# Verify indices exist
curl localhost:9200/_cat/indices?v
```

Services not healthy?

```
docker compose restart
docker compose logs --tail=100
```

Summary

Component	Purpose	Key Config
Filebeat	Ship logs from files	filebeat.yml
Logstash	Parse & enrich	logstash.conf
Elasticsearch	Store & index	docker-compose.yml
Kibana	Visualize	Index patterns

Next Steps: Explore the data, create custom visualizations, build your own dashboard!

Questions?

Lab Repository: `/home/morta/workspace/elk-lab`

Backup: File Structure

```
elk-lab/
├── docker/
│   ├── docker-compose.yml
│   ├── nginx.conf
│   ├── generate_ssh_logs.sh
│   ├── generate_syslog.sh
│   └── app/
├── config/
│   ├── filebeat.yml
│   └── logstash.conf
├── logs/
│   ├── nginx/
│   ├── ssh/
│   ├── syslog/
│   └── app/
└── docs/
    └── README.md
```